

STA630

Mid-Term (Solved)

ABSTRACT

This comprehensive collection of notes is accurately crafted to empower students to excel academically, ensuring they achieve a minimum of 80% marks in their examinations. The content is organized with clarity and precision, focusing on key concepts, critical analyses, and practical applications tailored to the syllabus. These notes serve as a reliable resource for both thorough preparation and last-minute revision. Designed to inspire confidence and mastery, this guide is an essential tool for students striving for academic excellence.

Maha Malik

RESEARCH METHODS

STA630_RESEARCH METHODS
Mid-Term (Solved)

Q. Why are basic research design issues important to consider before conducting the study and even during formulation of the research question?

Basic research design issues are crucial because they:

- Help clarify the research purpose.
- Determine the kind of data to be collected.
- Identify appropriate data collection methods.
- Prevent potential validity and reliability issues.
- Ensure resource optimization (time, cost, personnel).

Design decisions at the formulation stage guide the entire research process, from hypothesis development to conclusion, making the study scientifically sound and ethically feasible.

Q. Explain the two forms of scientific method and how both are important.

The two forms are:

- **Inductive Reasoning**: Moves from specific observations to general theories. For example, noticing repeated patterns and building a general conclusion.
- **Deductive Reasoning**: Moves from general theories to specific predictions. For example, testing a hypothesis derived from theory.

Both forms are interlinked in scientific research. Researchers often move back and forth between observation (induction) and theory (deduction) to refine and validate knowledge.

Q. Explain the ex post facto design.

In an **ex post facto design**, researchers study the effects of a variable after the fact—i.e., they do not manipulate variables but rather observe existing conditions or outcomes. It is used when manipulation is unethical or impossible. Researchers ensure objectivity by carefully selecting subjects to control extraneous factors.

Q. What do you understand by the term operationalization and how is it done?

Operationalization is the process of converting abstract concepts (e.g., morale, motivation) into measurable indicators. It involves:

- Defining the concept theoretically.
- Identifying dimensions and elements.
- Creating measurable variables or questions (e.g., Likert scales).

This allows empirical testing and statistical analysis.

Q. Internal vs External Validity – Lab vs Field Experiment

- **Lab Experiment**: High internal validity due to control over extraneous variables but low external validity because it's artificial.
- **Field Experiment**: High external validity (real-world setting), but lower internal validity due to lack of control.

Thus, a **trade-off** exists: to achieve high internal validity, researchers may sacrifice generalizability, and vice versa.

Q. Difference between Lab and Field Experiment

Feature	Lab Experiment	Field Experiment
Setting	Artificial/controlled	Natural/real-life
Control	High	Moderate to low
Validity Focus	Internal Validity	External Validity
Example	Testing memory in a quiet lab	Studying work performance on-site

Q. Difference between High Internal Validity and Low External Validity

- **High Internal Validity**: Ensures cause-effect inference (e.g., effect of X on Y). Focuses on **accuracy** within the study.
- **Low External Validity**: Limits generalizability beyond the study conditions.

Q. Six Data Collection Techniques

1. **Questionnaires**
2. **Interviews (structured/unstructured)**
3. **Observation**
4. **Experiments**
5. **Focus Groups**
6. **Content Analysis**

Q. What is Personal Interviewing? How is it conducted and where?

Personal interviewing involves face-to-face communication where the interviewer asks questions and records responses. It is conducted:

- At respondents' homes
- In offices
- On the street
- In shopping malls

It's useful for detailed responses but is costly and time-consuming.

STA630_RESEARCH METHODS
Mid-Term (Solved)

Q. Mail Questionnaires – Cost-effective but not Always

Mail questionnaires save costs on travel and manpower. However:

- Response rates can be low.
- No control over who completes it.
- Clarifications cannot be provided. Thus, effectiveness varies based on context and audience.

Q. Why Preliminary Notification is Important in a Self-Administered Questionnaire

It alerts respondents to expect the questionnaire, enhances credibility, and increases response rate. It may be sent as a postcard, email, or message.

Q. How Do Telephonic Interviews Reduce Interviewer Bias?

- No visual cues reduce stereotyping.
- Responses may be more candid due to perceived anonymity.
- Less impact from body language or facial expressions.

Q. "A Valid Instrument is Always Reliable, but a Reliable Instrument is Not Always Valid."

- **Reliability**: Consistency of results.
- **Validity**: Measures what it is supposed to measure. A scale may consistently measure weight (reliable), but if it's used to measure height, it's invalid.

Q. What is Test/Retest? How to Assess Reliability Using It?

Test/retest reliability involves:

- Administering the same test to the same group at two points in time.
- High correlation between results indicates good reliability.

Q. Difference between Test/Retest and Internal Consistency

- **Test/Retest**: Stability over time.
- **Internal Consistency**: Consistency within the test items (e.g., Cronbach's Alpha).

Q. Ways to Assess Reliability

1. **Test/Retest**
2. **Split-half method**
3. **Internal consistency (Cronbach's Alpha)**

4. **Inter-rater reliability**
5. **Parallel forms**
6. **Item-total correlation**

Q. "Literature Survey is Time-Consuming; In-depth Interview is Enough" – Discuss

Disagree. Literature surveys:

- Help in understanding prior research.
- Identify gaps and avoid duplication. In-depth interviews provide insights, but they cannot replace the broad scope of literature in building theoretical foundations.

Q. Resources for Conducting a Comprehensive Literature Review

- **Academic databases** (JSTOR, Google Scholar, ERIC)
- **University libraries**
- **Research journals**
- **Conference proceedings**
- **Books and dissertations**
- **Bibliographies of existing studies**

Q. What Do We Mean by Total Design Effort (TDE)? Why Is It Necessary?

TDE is a comprehensive approach to ensure high response rates and data quality. It includes:

- Preliminary contact
- Professional questionnaire design
- Follow-ups
- Incentives

TDE improves data reliability and study credibility.

Q. How Does a Research Question Help to Formulate a Hypothesis?

A research question identifies what is to be studied. From this, a testable hypothesis is developed. **Example**:
RQ: Does social media usage affect academic performance?
Hypothesis: Higher social media usage leads to lower academic performance.

Q. Longitudinal Research – Definition and Examples

Research repeated over time to track changes.
Examples:

STA630_RESEARCH METHODS
Mid-Term (Solved)

1. Studying employee satisfaction annually over 5 years.
2. Monitoring children's language development from age 2 to 10.

Q. Difference Between Confidentiality and Anonymity

- **Confidentiality**: Researcher knows identity but keeps it private.
- **Anonymity**: Even the researcher doesn't know the respondent's identity.

Q. Primary Causes of Job Stress

Common causes include:

- Work overload
- Role conflict and ambiguity
- Poor work-life balance
- Lack of support
- Job insecurity
- Poor communication

Q. What is Personal Interviewing, How It Is Conducted, and At Which Places?

Personal interviewing is a face-to-face method of data collection where an interviewer directly asks the respondent questions and records their answers.

Conducted at:

- Respondents' homes (door-to-door)
- Workplace
- Shopping malls (intercept interviews)
- Central interviewing facilities

Advantages:

- Higher response rate than mail/phone surveys
- Ability to observe non-verbal behavior
- Suitable for illiterate respondents
- Clarifications can be provided instantly

Disadvantages:

- Expensive
- Requires trained interviewers
- Less anonymity
- Difficult in high-security or remote areas.

STA630_RESEARCH METHODS
Mid-Term (Solved)

Q. Mail Questionnaires – Cost-Effective but Not Always

Mail questionnaires can be inexpensive because:

- No interviewer required
- Can reach geographically dispersed respondents

But they may not be cheap due to:

- Follow-ups increasing postage and printing costs
- Low response and completion rates
- Need for incentives and reminders to improve response

Hence, while initially low-cost, effective implementation may still require significant resources.

Q. Why Is Preliminary Notification Important in a Self-Administered Questionnaire?

Preliminary notification (via letter or phone):

- Increases response rates
- Prepares respondents for the survey
- Enhances credibility of the study
- Reduces the surprise element

Timing is crucial—notifications closer to the mailing time are more effective.

Q. How Telephonic Interviews Reduce Interviewer's Influence and Bias

- Lack of face-to-face contact reduces visual prejudice.
- Greater respondent anonymity.
- Can access difficult or sensitive demographics.
- Centralized control (e.g., CATI systems) improves standardization.

Q. What Is the Primary Difference Between Test/Retest and Internal Consistency?

- **Test/Retest**: Measures stability over time by repeating the test after a period.
- **Internal Consistency**: Assesses the extent to which items on the same test measure the same concept (e.g., using split-half or Cronbach's alpha).

Q. What Are the Different Ways of Assessing Reliability?

1. **Test-Retest Reliability**: Consistency over time
2. **Parallel-Form Reliability**: Correlation between different versions

3. **Split-Half Reliability**: Correlation between two halves of a test
4. **Inter-Item Consistency**: Correlation between items in a scale
5. **Inter-Rater Reliability**: Consistency between different raters
6. **Internal Consistency (Cronbach's Alpha)**.

Q. "Literature Survey Is Time-Consuming; In-Depth Interview Is Sufficient." Discuss

Disagree:

- **Literature survey** provides comprehensive background, identifies gaps, and avoids duplication.
- In-depth interviews offer insight but lack breadth.
- A strong theoretical framework requires both data and context. Thus, relying only on interviews limits the depth and credibility of research.

Q. Resources for Conducting a Comprehensive Literature Review

- **University libraries and digital repositories**
- **Google Scholar, JSTOR, ScienceDirect**
- **ERIC, Springer, PubMed** (if relevant)
- **ResearchGate, Academia.edu**
- **Bibliographies of key research articles**
- **Conference proceedings and dissertations**

Q. What Do We Mean by Total Design Effort (TDE)? Why Is It Necessary?

TDE refers to using a combination of techniques to maximize response rate and data quality, including:

- Preliminary notifications
- Personalized cover letters
- Incentives (money, gifts)
- Follow-ups
- Attractive questionnaire layout

It's necessary because no single technique is sufficient to ensure high response rates. A total effort ensures better reliability and validity of survey results.

Q. How Does a Research Question Help to Formulate a Hypothesis?

A clear research question identifies the focus of the study. From this, a hypothesis is created as a testable statement.

Example:
Research Question: Does social media usage affect sleep quality among university students?
Hypothesis: Increased social media usage negatively affects sleep quality.

This enables focused testing and analysis.

Q. Give Two Examples of Longitudinal Research

1. **Health tracking**: Monitoring diabetic patients' health over a 10-year period.
2. **Educational outcomes**: Studying a cohort of students from primary school through college to observe academic progression and career success.

These studies track changes over time and are ideal for developmental and behavioral research.

Q. Differentiate Between Confidentiality and Anonymity

- **Confidentiality**: Researcher knows the participant's identity but keeps it private.
- **Anonymity**: Even the researcher does not know the respondent's identity.

Confidentiality protects data; anonymity protects identity.

Q. Primary Causes of Job Stress

1. Work overload or underload
2. Role conflict and ambiguity
3. Job insecurity
4. Poor management or lack of support
5. Interpersonal conflicts
6. Lack of work-life balance
7. Unclear expectations or goals

These stressors impact employee health and performance.

B.Ed. (1.5 Years) BS English Files